Towards Trustworthy Retrieval Augmented Generation for Large Language Models: A Survey

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ACM 1557-7341/2026/8-ART
https://doi.org/10.1145/3837074

Retrieval-Augmented Generation (RAG) enhances AI-generated content by integrating external knowledge, improving relevance, and reducing hallucinations. However, RAG also introduces risks related to reliability, safety, privacy, fairness, explainability, and accountability, which impact trustworthiness. While various methods aim to address these concerns, a unified framework is lacking. This survey bridges that gap by presenting a comprehensive roadmap for trustworthy RAG systems. We provide a structured analysis of key challenges, existing solutions, and future directions across these aspects. We further organize the field around how the retrieval and generation stages and the six trustworthiness dimensions interact, demonstrating their synergies and trade-offs through a controlled cross-dimensional analysis. Additionally, we highlight downstream applications where trustworthy RAG can make a significant impact, encouraging further research and adoption in real-world AI systems.

CCS Concepts: • Computing methodologies → Artificial intelligence; Natural language processing.

Additional Key Words and Phrases: Accountability, Explainability, Fairness, Large Language Models, Privacy, Reliability, Retrieval-Augmented Generation, Safety, Trustworthy AI

1 Introduction

Retrieval-Augmented Generation (RAG) has emerged as a promising approach to address key limitations of Large Language Models (LLMs), such as hallucinations, outdated knowledge, and limited explainability [45]. By incorporating external information into the generation context, RAG improves accuracy, reliability, and keeps models up-to-date with minimal retraining. These benefits have profound implications for real-world applications, such as effectively being applied in medical question answering [171], legal document drafting [116, 168], and educational chatbots [151].

In 2022, the National Institute of Standards and Technology (NIST) published guidelines for trustworthy AI, defining trustworthiness across six dimensions: Reliability, Privacy, Safety, Fairness, Explainability, and Accountability [148]. While these principles apply broadly to AI systems, the multi-stage retrieval-augmentation-generation pipeline introduces unique challenges that demand RAG-specific reinterpretations of each dimension, as we elaborate below.

Reliability in general AI concerns consistent and accurate performance under varying conditions. In RAG systems, reliability must be evaluated across two interdependent stages: retrieval, where inconsistent or irrelevant passage retrieval can propagate errors downstream, and generation, where the model must faithfully ground its outputs in retrieved content rather than hallucinate [45, 80].

Privacy in general AI focuses on safeguarding personal data from unauthorized access. In RAG systems, privacy risks extend beyond model memorization to corpus-level exposure: the integration of external databases introduces additional leakage channels through which adversaries can extract sensitive documents via carefully crafted queries a threat absent in standalone LLMs [188]. For example, a healthcare RAG assistant must guard against both direct extraction of patient records from its knowledge base and the inadvertent surfacing of sensitive details in generated responses.

Safety in general AI addresses harm prevention and robustness against adversarial actors. In RAG systems, the retrieval corpus itself becomes a first-class attack surface: malicious content injected into the knowledge base can hijack the generation stage, producing misinformation or harmful outputs even

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when the underlying LLM is otherwise well-aligned [37, 174, 187]. This risk is distinct from conventional jailbreaking threats and requires dedicated corpus integrity safeguards, particularly in high-stakes deployments such as medication advising or legal consulting.

Fairness in general AI targets the mitigation of demographic biases in model outputs. In RAG systems, bias can compound across two stages: retrieval bias, where certain groups are underrepresented in retrieved passages, and generation bias, where the LLM amplifies these disparities in its final response [138]. Critically, retrieved content can inflate the model's confidence in biased outputs, making RAG-induced unfairness harder to detect than bias in standalone LLMs [58].

Explainability in general AI concerns the transparency of model decision-making. In RAG systems, explainability requires attribution across both retrieval and generation: users must understand not only what the system generated, but which retrieved passages caused it and why those passages were selected. This dual attribution requirement goes beyond token-level explanations sufficient for standalone LLMs, and remains a significant open challenge as retrieved information often influences generated responses in opaque ways.

Accountability in general AI encompasses governance, audit trails, and traceability of system decisions. In RAG systems, accountability is further complicated by the distributed pipeline: responsibility is shared among the retriever, the knowledge base curator, and the generator, creating accountability gaps when errors or harms arise.

Due to the importance of trustworthiness, a plethora of research has been developed to advance the application of RAG in Large Language Models. However, there is no systematic review of this area's current advancements and challenges. To organize the various perspectives, this survey formulates a systematic discussion on the state of trustworthy RAG in Large Language Models. Moreover, we recognize that trustworthiness in RAG cannot be understood one dimension at a time. RAG is a multi-stage pipeline where trustworthiness aspects are coupled across stages and dimensions; thus, an intervention targeting one property might affect another. We therefore structure our analysis around these interactions in Section 9 with a controlled cross-dimensional experiment, demonstrating the synergy and trade-offs across trustworthiness interventions. Finally, in Section 10, we ground these dimensions in high-stakes deployment domains, namely healthcare, legal, education, and enterprise, where we revisit how each trustworthiness dimension manifests in practice. The list of papers discussed is provided in the GitHub repository[1].

2 Preliminaries

This section introduces the RAG framework for LLMs and its common downstream tasks. While RAG also applies beyond language (e.g., image generation), we limit our scope to RAG in LLMs, sometimes called Retrieval Augmented Language Models (RA-LLMs) [36], and use RAG, RA-LLM, and RAG-based systems interchangeably.

2.1 Retrieval Augmented Generation

Figure 1 overviews the trustworthy RAG framework: a central three-stage pipeline (retrieval, knowledge augmentation, and generation) surrounded by the six trustworthiness dimensions examined in this survey, namely Reliability, Privacy, Safety, Fairness, Explainability, and Accountability, each shown with the key challenges and mechanisms that arise as it interacts with the pipeline.

[1] https://github.com/Arstanley/Awesome-Trustworthy-Retrieval-Augmented-Generation

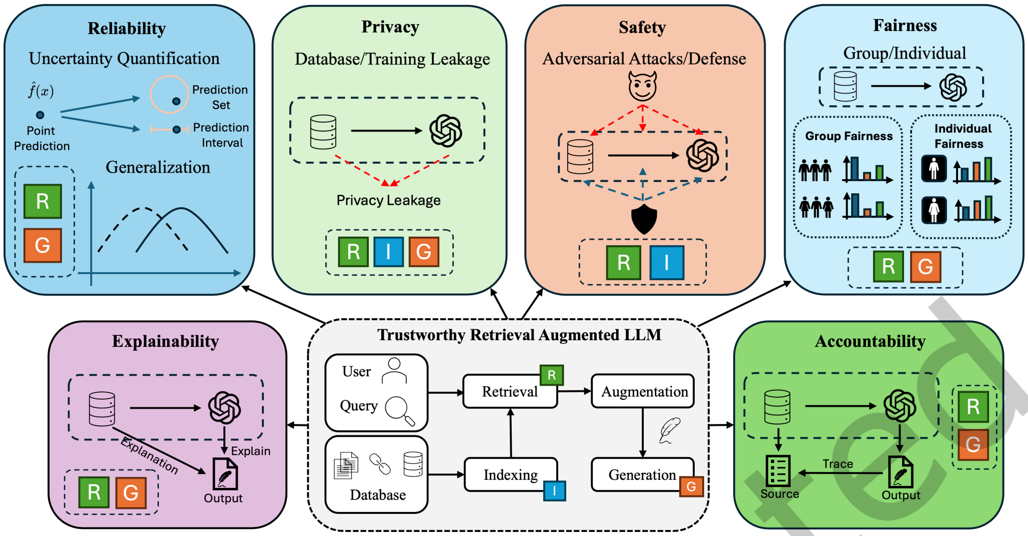


Fig. 1. An overview of the key components and dimensions of Trustworthy Retrieval Augmented Generation (RAG) for Large Language Models (LLMs) that are covered in this survey.

Dense Retrieval. Given a query, the system first retrieves relevant context. Following [45], retrieval comprises indexing, where heterogeneous sources (e.g., PDFs, HTML, Markdown) are chunked, embedded, and stored in a vector database, or a symbolic or relational database for tasks such as knowledge graphs (e.g., GraphRAG), and retrieving, where the query is vectorized or aligned to symbolic structures to fetch relevant chunks or nodes. Knowledge augmentation then incorporates the retrieved content into generation, often via prompt injection or fine-tuning, and the generation stage produces the final response.

Agentic Retrieval. Recently, agentic retrieval has emerged as a new paradigm in which an LLM-based agent decides when, what, and how to retrieve and integrate information across multiple reasoning steps [137, 182]. ReAct [182] merges reasoning traces with retrieval actions to refine queries from intermediate results; Reflexion [137] adds self-reflection on past failures; and Self-RAG [11] trains models to decide when retrieval is needed and to critique retrieved passages.

Scope of This Survey. We note that while this survey introduces both dense and agentic retrieval paradigms, the technical coverage in Sections 3 through 8 focuses primarily on dense retrieval settings, as the majority of existing trustworthy RAG research has been conducted in this context. Agentic retrieval introduces additional trustworthiness challenges beyond those of dense retrieval, such as error propagation across multi-step reasoning chains and the difficulty of attributing failures to specific retrieval decisions, and we identify these as important open directions for future work.

2.2 Common Tasks in RAG

The RAG paradigm has enabled a wide range of applications. We categorize the most common downstream tasks into two categories based on their interaction pattern: interactive tasks, where the system returns a response conditioned on a user query, and generative tasks, where the system produces a standalone artifact rather than a reply.

Interactive Tasks. In interactive tasks the system answers a user query, either in a single turn or across multiple turns. Question Answering (QA) is one of the primary downstream tasks for RAG-based

Table 1. Comparison with Existing Surveys on RAG and Trustworthy LLMs. Column headers are colored by the meaning of trustworthiness each sub-dimension primarily engages (cf. §2.4): **M1** (trust challenges introduced by RAG) and **M2** (RAG as a mechanism to improve trust).

		Pillars of Trustworthiness											
		Reliability		Privacy		Safety		Fairness		Explainability		Accountability	
Surveys		Uncertainty	Generalizability	External	Training Data	Jailbreaking	Defense	Retrieval	Generation	Retrieval	Generation	Retrieval	Generation
LLM	[87]	$\not\checkmark$	$\not\checkmark$	✗	$\not\checkmark$	$\not\checkmark$	$\not\checkmark$	✗	$\not\checkmark$	✗	$\not\checkmark$	✗	$\not\checkmark$
	[60]	✗	✗	✗	$\not\checkmark$	$\not\checkmark$	$\not\checkmark$	✗	$\not\checkmark$	✗	✓	✗	$\not\checkmark$
RAG	[36]	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗
	[45]	✗	✓	✗	✗	✗	✓	✗	✗	✓	✓	✗	✗
	[197]	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗
	[106]	✗	✗	✓	✓	◐	◐	◐	◐	◐	✗	✗	✗
	[159]	✓	✓	✓	◐	✓	✓	◐	◐	◐	◐	◐	◐
Ours		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

$\not\checkmark$: LLM-focused survey without RAG emphasis; ◐: shallow coverage (e.g., mention in the future direction without dedicated discussion).

language models, where the system generates accurate answers to user queries using external retrieved context [93, 144]. Dialogue extends QA to multi-turn task-oriented and open-domain conversation, integrating external knowledge for more informed and up-to-date responses [4, 38].

Generative Tasks. In generative tasks the output is a produced artifact rather than a reply within an exchange. Code Generation retrieves documentation, code snippets, and issue histories to support program synthesis [22, 34]. Beyond code, RAG-based language models support generative recommendation [79], software engineering [57], and AI for scientific discovery [3].

Trustworthy Evaluation. Comprehensive trustworthiness assessment requires metrics that capture bias, fairness, and reliability in generated answers. We introduce dimension-specific metrics in detail throughout the survey in their corresponding sections.

2.3 Motivation

Although trustworthiness has been studied in standalone LLMs, RAG systems introduce unique challenges. Their multi-stage pipelineindexing, retrieval, and generationcan compound issues like hallucinations and bias, while reliance on external sources increases susceptibility to unreliable information. These distinctions limit existing framework utility. While prior works address individual concerns [33, 176, 188, 197], a unified overview is missing. This survey addresses that gap by systematically reviewing current efforts and outlining future directions for trustworthy RAG.

Related Surveys and Differences. As shown in Table 1, existing RAG surveys [36, 45, 106, 159] primarily focus on architectures, system design, and benchmarks, offering limited analysis of trustworthiness. Recent work discusses trust-related issues from architectural or system perspectives [159] or focuses on trustworthy empirical evaluations [197]. Meanwhile, trustworthy LLM surveys [60, 87] overlook retrieval-specific issues. In contrast, our survey provides a systematic review of RAG through six trustworthiness dimensions: Reliability, Privacy, Safety, Fairness, Explainability, and Accountability.

Paper Collection Methodology. We followed a systematic literature review methodology, searching Google Scholar, ACM Digital Library, and arXiv using combinations of primary terms (e.g., "Retrieval-Augmented Generation," "trustworthiness") and dimension-specific terms (e.g., "RAG bias," "RAG privacy attack"). Papers were included if they addressed trustworthiness challenges or evaluation protocols within RAG-based LLM systems. Forward and backward citation chasing was applied to recover works missed by keyword search. The search cutoff is July 2025, yielding a final corpus of nearly 196 papers. Our online repository is actively maintained with up-to-date papers.

2.4 Trustworthiness Definition

We note that the literature uses the notion of RAG trustworthiness in two distinct senses, which we make explicit to avoid ambiguity. First (M1), RAG introduces trust challenges that do not arise for a standalone generative model by including new attack surfaces including an external database and retrieval mechanisms. Second (M2), RAG can serve as a mechanism for improving trust, by directly grounding outputs in retrieved evidence or producing faithful explanations. Table 1 maps each sub-dimension to the meaning it primarily engages through color, so the reader can situate each subsequent section accordingly.

3 Reliability of Retrieval Augmented Generation

While RAG improves factual consistency and adaptability, it also introduces unique reliability challenges. Unlike standalone generative models, RAG reliability depends not only on the underlying LLM but also the alignment of retrieved information. Ensuring reliability in RAG therefore requires evaluating both the retrieval process and the generation conditioned on the retrieved content.

3.1 Taxonomy of RAG Reliability

At a high level, reliability requires the system to perform as expected under various conditions. Previous work [155] defines reliability from three aspects: the ability to express uncertainty in predictions, the capability of robust generalization under various conditions, and the extent to which the model can adapt to new tasks. Since RAG is inherently adaptable because of the retrieved context, we will only consider uncertainty and robust generalization in our following discussion. We note that this taxonomy reflects one principled organizational choice; papers addressing multiple sub-dimensions simultaneously could reasonably be categorized differently, and we acknowledge this applies to the taxonomies presented throughout the survey.

3.2 Uncertainty

Uncertainty is a crucial factor for model reliability. Uncertainty quantification (UQ) helps quantify the confidence in the model's predictions, which is essential in high-stakes scenarios. Consider a medical chatbot where a patient inquires about their condition. If the model can express uncertainty in its responses, it significantly reduces the risk associated with its predictions. The patient can then make more informed judgments based on the confidence level of the information provided.

For a RAG system, uncertainty quantification presents two primary challenges: First, during the generation phase, uncertainty stems from the inherent limitations of LLMs. Standard techniques for quantifying uncertainty in LLMs, such as conformal prediction, can be applied here with few adaptations [183]. Second, uncertainty arises during the retrieval phase and its interaction with the LLM, introducing a more complex dynamics. The combination of retrieval and generation processes creates unique challenges for UQ, necessitating advanced methods to address the overall system complexity. The following section outlines ongoing efforts to tackle these challenges, with a summary of the relevant literature presented in Table 2. The Scope column of marks whether each method is RAG-specific (RAG) or a general LLM technique applicable to RAG's generator (LLM). Although LLM-specific methods can be directly applied to the downstream generation module, they address only the first challenge above; they remain agnostic to the retrieval stage and lack the joint calibration across retrieval and generation that the second, RAG-specific challenge demands. This scope distinction will be adopted throughout the survey for taxonomy tables.

Table 2. Taxonomy for Uncertainty Quantification in RAG. The Scope column distinguishes methods designed for or evaluated specifically on RAG (RAG) from general LLM techniques that are applicable to RAG's generator but not RAG-specific (LLM).

Module	Reference	Scope	White-box	Task	Year
Generation	Ye et al. [183]	LLM	7	Benchmarking	2024
	Su et al. [142]	LLM	✓	Open Domain Question Answering	2024
	Kumar et al. [74]	LLM	✓	Multiple Choice Question Answering	2023
	Quach et al. [123]	LLM	7	Open Domain Question Answering	2023
	Wang et al. [165]	LLM	7	Open Domain Question Answering	2025
Retrieval + Generation	Ni et al. [104]	RAG	7	Multi-hop Question Answering	2024
	Li et al. [80]	RAG	7	Open Domain Question Answering	2023
	Soudani et al. [141]	RAG	7	Open Domain Question Answering	2026

3.2.1 Uncertainty Quantification in Generation. The generation phase in a RAG system is critically influenced by the UQ of LLMs. Recent studies have explored various approaches in this area, with a focus on techniques like conformal prediction (CP) a model-agnostic, distribution-free method that uses a calibration set to estimate prediction confidence [135]. To apply conformal prediction, a non-conformity score is first defined to measure the confidence of a given prediction. Using a calibration set, the $1-\alpha$ quantile of the non-conformity score is then calculated, where α represents the user-defined error rate. Finally, the prediction set is constructed by selecting valid predictions based on the quantile score, ensuring that the set satisfies the $1-\alpha$ confidence level, assuming the calibration and test sets are exchangeable. Beyond conformal prediction, complementary approaches have explored confidence calibration and verbalized confidence elicitation for LLMs [46, 172], which estimate model reliability through calibration techniques such as temperature scaling or by prompting models to self-assess their certainty; however, adapting these methods to the multi-component RAG pipeline remains an open challenge.

The cornerstone of CP lies in defining the non-conformity score. In traditional multi-class classification, a common approach is to use the softmax score of the from the class prediction. Extending the logit-based non-conformity score to LLMs, methods have been further developed. Typically, they assume white-box access to the model, making them unsuitable for commercial LLMs such as ChatGPT. For instance, Kumar et al. [74] applied standard CP to the Llama model [154] by leveraging softmax scores of token logits in multiple-choice tasks. Similarly, Ye et al. [183] extended logit-based approaches to multiple baselines and language models.

To compensate for the lack of application on black models, another promising direction is proposed for sampling-based techniques, where model confidence is estimated by repeatedly prompting the LLM. Quach et al. [123] adapted the learn-then-test risk-control framework [9] for LLMs, approximating the non-conformity score through sampling, which allows uncertainty quantification in black-box models without direct logit access. Su et al. [142] further advanced these methods by introducing non-conformity measures that integrate both coarse-grained and fine-grained notions of uncertainty, leading to smaller, more refined prediction sets. Beyond constructing prediction sets, Wang et al. [165] propose COIN, a modular calibration framework that uses confidence interval methods such as ClopperPearson to establish high-probability upper bounds on the true error rate, enabling threshold-based selective prediction with provable false discovery rate guarantees under both white-box and black-box settings.

3.2.2 Uncertainty Quantification in Retrieval and Generation. As shown in Figure 1, a traditional RAG system includes multiple components from retrieval to generation. Due to its complex, multi-component nature, directly applying LLM-based UQ methods will produce less accurate, sub-optimal results [80, 130]. This necessitates the development of specialized techniques tailored to the unique structure and requirements of RAG models.

Recently, researchers proposed a multi-step calibration framework to enhance the retrieval process of RAG [130]. Specifically, this framework uses conformal prediction to quantify retrieval uncertainty, ensuring trustworthiness in RAG systems. The framework involves constructing a calibration set of questions answerable from the knowledge base and comparing their embeddings against document embeddings to identify the most relevant chunks containing the answers. By analyzing similarity scores and determining a cutoff threshold based on a user-specified error rate (α), the system retrieves all chunks exceeding this threshold during inference. This multi-step calibration ensures the true answer is captured in the context with a (1 - α) confidence level.

Moreover, TRAQ [80] expanded the conformal prediction framework to include a Bayesian optimization module that minimizes the prediction set during the multi-step calibration. Because of the complexity of RAG, the constructed prediction set will be very large after aggregating the error rates of multiple components. By leveraging Bayesian optimization, the framework efficiently searches for the optimal parameters that reduce the size of the prediction set while maintaining the desired confidence level. TRAQ ensures that the retrieval process remains both accurate and computationally feasible, enhancing the overall reliability and performance of RAG systems.

Besides uncertainty quantification in the process of document retrieval, Uncertainty-Aware Graph Reasoning (UaG) [104] attempted to address the gap in uncertainty quantification of knowledge graph reasoning. Unlike vector databases, knowledge graphs encode knowledge as triplets and include structural information. One representative task of knowledge graph reasoning is multi-hop question answering, where the system must infer answers by traversing multiple edges in the knowledge graph. The UaG framework involves combining information from several related entities and relationships within the graph, further complicating the process of uncertainty quantification. UaG used a general risk control framework to find the optimal parameter for each stage of calibration, ensuring reliable uncertainty estimates while maintaining a reasonable prediction set size. Soudani et al. [141] further extended UQ to multi-step retrieval-augmented reasoning by perturbing reasoning chains to capture uncertainty propagated across both retrieval and generation.

3.3 Uncertainty Evaluation

Metrics. Traditionally, uncertainty quantification is evaluated from two key perspectives: coverage and efficiency [55]. Recall that the goal of uncertainty quantification is to ensure that the returned answer set satisfies a user-defined error tolerance of $1 - \alpha$. Thus, the coverage rate measures how effectively the model meets this requirement.

Given a returned set of answers, $\mathcal{A}_{\text{ret}}$, and the correct answer set, $\mathcal{A}_{\text{true}}$, the coverage rate, C, is calculated as the proportion of instances where the correct answer is included in the returned set. Formally, $C = \frac{N_{\text{correct}}}{N_{\text{total}}}$, where N_{correct} represents the number of times the correct answer $\mathcal{A}_{\text{true}}$ is contained in the returned set $\mathcal{A}_{\text{ret}}$, $\mathcal{A}_{\text{true}} \subseteq \mathcal{A}_{\text{ret}}$, and N_{total} is the total number of instances. A model is considered reliable when $C \geq 1 - \alpha$. However, simply exceeding the threshold does not necessarily indicate optimal performance. Overestimating the returned set size while still satisfying the desired error rate implies inefficiency, as a smaller set could suffice for the same error rate. As a result, alongside coverage, efficiency, denoted as E, is another critical metric, often evaluated by the size of the returned answer set (i.e. the number of returned answers per question), i.e., $E = |\mathcal{A}_{\text{ret}}|$. Efficiency reflects the utility of the models output, as larger sets may contain more irrelevant information, reducing their usefulness to the user. Thus, an efficient uncertainty quantification process minimizes E while maintaining the desired coverage rate, C.

3.4 Robust Generalization

Previous work [155] defines robustness as the ability to make accurate estimates or forecasts about unseen events caused by out-of-distribution data, covariate shift, domain change, concept change, or population shift. In the context of RAG, the most significant robustness challenge is that the knowledge base itself is unreliable: as the database evolves it accumulates spurious, outdated, or factually incorrect content, so that even an accurate retriever returns harmful context. This is distinct from the standard retrieval-relevance problem in which an imperfect retriever surfaces off-topic passages from an otherwise clean corpus. We therefore focus on noise that originates in the corpus and arises organically as the database evolves, and we consider two aspects, distinguished by the quality of the offending content: resilience against irrelevant context, namely passages unrelated or only superficially related to the query, and resilience against corrupted context, namely passages carrying factually incorrect information. We further note a third type, adversarially constructed corrupted context; because of its adversarial nature we treat it under adversarial robustness in Section 5 (Safety).

3.4.1 Irrelevant Context. Fang et al. [37] considers the noise robustness of RAG with adaptive adversarial training. The paper explores three types of retrieval noises: (i) contexts that appear to be related to the query but do not contain the correct answer, (ii) contexts that are entirely unrelated to the query, and (iii) contexts that are thematically related to the query but include incorrect information. With the conclusion that type (i) and (iii) noise are the most misleading to the language models, the authors developed Retrieval-augmented Adaptive Adversarial Training (RAAT) to regulate the retrieval of noisy text. To improve the robustness under the noisy data, RAAT generates adversarial samples (noises) by considering the model's sensitivity to various types of noises and shows significant robustness improvement. The study further demonstrates that RAAT can be integrated seamlessly with existing RAG systems, enhancing their performance without substantial computational overhead.

In addition, Yoran et al. [187] further explores the negative impact of the retrieval of irrelevant context on the model performance. They argue that the negative impact of the irrelevant context is a result of the lack of training data with the retrieved passages. As a result, the brittleness to noisy passages is expected during inference. To address this observation, the author propose to finetune the language models on noisy contexts, which allows the model to learn to differentiate between useful and distracting information and minimize the negative effect of irrelevant context. Their experiment result on five open domain datasets has shown significant improvements of robustness against irrelevant context for both single-hop and multi-hop retrieval based question answering.

3.4.2 Corrupted Context. Recently, Xu et al. [174] proposed a theoretical framework to explore the benefits and detriments of the RAG, in the situation where there's a discrepancy between the retrieved knowledge and the LLM knowledge. Specifically, they observed that the similarity between the RAG representation and the retrieved representation is bounded by the benefits and detriments, and the similarity is positively correlated with the value of benefits minus detriments. These results suggest that the similarities can be used as a proxy for the benefits and detriments of the RAG. Building upon the theoretical results, they further proposed an interactive inference framework X-RAG that leverages the benefit of both retrieved knowledge and LLM knowledge.

3.5 Robustness Evaluation

Metrics and Datasets. Robustness evaluation applies standard QA metrics (see Section 2.2) to noisy retrieval settings, with results reported as absolute performance tables [37, 174] or performance deltas relative to a clean baseline [187]. However, no widely adopted benchmark exists, so they adapt common

Table 3. Taxonomy for RAG Privacy

	Reference	Training	Tasks	Leakage	Year
Attack	Zeng et al. [188]	✓	Document Extraction & Training Data	Internal & External	2024
	Liu et al. [85]	✓	Membership Inference Attack	External	2024
	Cohen et al. [26]	7	MIA & Document Extraction	External	2024
	Jiang et al. [65]	7	IP Leakage (CopyBreakRAG)	External	2024
	Peng et al. [113]	✓	Document Extraction	External	2024
	Chaudhari et al. [162]	7	IP Leakage (Benchmark)	External	2025
Defense	Zeng et al. [189]	✓	External Database	External	2024
	Zeng et al. [188]	✓	Document Extraction	External	2024
	Fu et al. [43]	7	Membership Inference Attack	External	2026

QA benchmarks (e.g., TriviaQA) by injecting either randomly selected or heuristically filtered noise depending on whether the setting targets irrelevant or corrupted context [174, 187]. A recent step toward standardization is the benchmark of Fang et al. [37], which provides three types of retrieval noise per instance—relevant, irrelevant, and counterfactual—alongside a golden retrieval document, and we encourage future works to adopt it for standardized comparison.

3.6 Future Directions of RAG Reliability

Unified Uncertainty and Robustness Framework. Current approaches treat uncertainty quantification and robust generalization as separate problems, yet in practice they are tightly coupled noisy retrieved passages increase uncertainty, while miscalibrated confidence amplifies the impact of corrupted context. A concrete direction is to develop a joint framework that dynamically adjusts confidence scores based on retrieval quality signals, enabling RAG systems to simultaneously resist noise and communicate calibrated uncertainty to users. The discussion in Section 9 further demonstrates the potential synergy between reliability and robustness.

Adaptive Reliability under Knowledge Drift. External knowledge bases often evolve over time, yet RAG systems currently lack mechanisms to detect and adapt to such shifts. Drawing on advances in dynamic knowledge graphs [104, 145] and active learning, a promising direction is to develop distribution shift aware intervention mechanisms that flag outdated passages and trigger uncertainty escalation when retrieved content conflicts with model priors.

4 Privacy of Retrieval Augmented Generation

Although privacy risks in LLMs are well-studied, integrating external data in RAG introduces additional challenges. This section introduces the threat model for privacy leaks in RAG, discusses current mitigation efforts, and explores future directions for enhancing RAG trustworthiness.

4.1 Taxonomy of RAG Privacy

Table 3 summarizes work addressing privacy in RAG; we describe the relevant taxonomy below. Unlike uncertainty quantification, privacy in RAG has been studied almost exclusively in the retrieval-augmented setting; we therefore omit the scope annotation for this section, as all surveyed works are RAG-specific.

4.1.1 Training. Training indicates whether attacks or defenses require prior data training. Approaches requiring training usually assume white-box access, allowing fine-tuning of RAG components. In contrast, methods without training often rely on prompt-based or zero-shot techniques.

4.1.2 Tasks. Current RAG privacy research involves four tasks: Document Extraction, Membership Inference, Intellectual Property Leakage, and Training Data Extraction. Document Extraction aims to retrieve confidential data (e.g., Personally Identifiable Information) from external databases. Membership Inference determines if specific passages exist in databases, potentially exposing sensitive data. Intellectual Property Leakage represents a distinct and higher-order threat: rather than targeting individual PII fields, adversaries aim to exfiltrate entire proprietary documents such as contracts or trade secrets. Lastly, Training Data Extraction explores leakage of sensitive information from the LLMs internal training data.

4.1.3 Leakage. Privacy leakage can originate from two sources: external retrieval databases and internal training data. External leakage involves exposing sensitive information from retrieved knowledge sources, while internal leakage occurs when LLMs inadvertently reproduce confidential training data. The following discussion is structured around these two aspects. These two aspects engage different senses of RAG trustworthiness in the sense of Section 2.4: external-database leakage is a risk that RAG introduces (M1), whereas the training-data setting illustrates how RAG can instead reduce leakage (M2).

4.2 Data Leakage From the External Retrieval Database

The goal of the attacker is to exploit privacy vulnerabilities within the retrieval dataset. Zeng et al. [188] introduced a composite structured prompt, formulated as q = information + command, which leverages the context retriever's propensity for similarity-based matching. However, a significant limitation of this approach is its reliance on fixed queries, which cannot dynamically adapt to varying contexts. To address this limitation, Jiang et al. [65] proposed CopyBreakRAG, an agent-based black-box attack that begins with an initial adversarial query and iteratively refines it based on the model's responses, dynamically switching between curiosity-driven exploration and reasoning-based exploitation to progressively extract as many documents as possible from the retrieval database. Compared against PIDE [120], a structured prompt injection baseline, and DGEA [26], a grey-box method requiring direct access to the embedding model, CopyBreakRAG outperforms PIDE by approximately 45% and DGEA by approximately 25% in chunk recovery rate under fully black-box conditions.

When considering white-box access to the model, Peng et al. [113] focused on data extraction through backdoor attacks. Their method trains a model to associate specific triggers with desired outputs. Beyond directly extracting documents, their approach also explores generating stealthy outputs using a language model to paraphrase the retrieved content, thereby increasing the difficulty of detecting the attack. Furthermore, Cohen et al. [26] demonstrated that these attacks can escalate beyond isolated cases. By crafting an adversarial self-replicating prompt, attackers can initiate a chain reaction that propagates through the entire RAG system.

Although distinct from direct extraction methods and based on a different threat model, membership inference attacks have also proven effective for data extraction. These attacks allow malicious users to infer whether specific content is present in the retrieval database. Liu et al. [85] introduced a mask-based attack that obscures portions of documents, compelling the language model to predict the masked words. Fu et al. [43] systematically compared the vulnerability of RAG and SFT against seven representative MIAs under a unified protocol, finding that RAG is generally more resistant since knowledge resides in an external index rather than in updated parameters.

4.3 Data Leakage From the LLM Training Data

The goal of the attacker is to extract data from the LLM's training and fine-tuning data that are encoded in the model parameters. In their paper, Zeng et al. [188] compared the effect of RAG in preventing data leakage from the LLM training data. The result shows that incorporating retrieved passages greatly reduces LLM's propensity to reproduce content memorized during its training/fine-tuning process. To isolate the effect of retrieval data integration, the author also attached 50 tokens of random noise injection as prefix. Although the random noise could also mitigate the data leakage, it is far less effective than integrating the retrieved content.

4.4 Intellectual Property Leakage

Beyond PII extraction, another privacy challenge is the extraction of proprietary documents from the retrieval corpus. Unlike PII attacks that target specific sensitive fields, IP leakage aims to reconstruct the knowledge base, enabling adversaries to obtain confidential organizational assets such as contracts and trade secrets. CopyBreakRAG [65] demonstrates this threat concretely, introducing an agent-based black-box attack that iteratively refines adversarial queries based on system feedback. Cohen et al. [26] further show that self-replicating prompts can escalate such attacks across an entire RAG system. A recent benchmark [162] confirms that IP leakage generalizes across diverse retriever and generator architectures, further emphasizing the need for dedicated defenses beyond PII extraction.

4.5 Defense on Privacy Attacks

Although still a relatively under-explored area, some works have proposed defenses to mitigate privacy vulnerabilities in RAG systems. In their foundational work, Zeng et al. [188] observed that using a separate model to summarize the retrieved documents effectively reduces privacy leakage by abstracting sensitive information into generalized content. Additionally, they proposed implementing a distance threshold in the retrieval database, ensuring that only documents with certain relevance requirements are returned. However, this approach introduces a trade-off between system performance and privacy protection, as stricter thresholds can limit retrieval accuracy.

Building on these mitigation strategies, the authors further suggested the use of purely synthetic data as a way to entirely avoid potential leakage of real data [189]. This method involves identifying important attributes of the data through few-shot samples, extracting key information associated with these attributes, and generating synthetic data that mirrors the original data without exposing sensitive information. This approach has shown promise in effectively mitigating privacy leakage while maintaining the performance of the RAG system. Additionally, recent research [43] has explored LLMs as a post-hoc privacy mitigation for MIA attacks.

Metrics and Datasets. Privacy evaluation metrics quantify information leakage via token overlap volume, unique excerpts extracted, and ROUGE-L similarity between extracted and original content [188, 189]. For IP leakage specifically, Chunk Recovery Rate (CRR), Semantic Similarity (SS), and Extended Edit Distance (EED) jointly assess verbatim reconstruction fidelity at the corpus level [65]. However, no standardized benchmark currently exists; commonly used datasets include the Enron Email dataset [72], HealthcareMagic-101 [189], and domain-specific corpora simulating proprietary knowledge bases [65].

4.6 Future Directions of RAG Privacy

Domain-Specific Privacy Mechanisms. Current privacy defenses are evaluated on general-domain corpora, yet threat profiles sometimes differ substantially across domains. For example, re-identification from clinical notes in healthcare and leakage of privileged communications in law can have different implications in terms of model design. Future work should develop domain-adapted anonymization pipelines with sensitivity-based redaction at indexing time, evaluated on domain-specific benchmarks such as MedicationQA [15] and LegalBench-RAG [116].

Table 4. Taxonomy for RAG Safety

	Reference	White-Box	Black-Box	Year
Targeted	Zou et al. [198]	✓	✓	2024
	Xue et al. [176]	✓	7	2024
	Long et al. [89]	✓	7	2024
	Zhong et al. [196]	✓	7	2023
	Chang et al. [20]	✓	7	2025
	Zhao et al. [194]	✓	7	2025
Jailbreak	Wang et al. [166]	7	✓	2024
	Deng et al. [32]	7	✓	2024
	An et al. [7]	7	✓	2025

Unified Privacy Evaluation Benchmark. Current evaluations focus narrowly on PII extraction, leaving IP leakage [65], membership inference attacks [43] and corpus reconstruction attacks undercharacterized. A promising direction is a unified RAG privacy benchmark covering these threat categories: direct extraction, PII inference, IP leakage, and membership inference.

5 Safety of Retrieval Augmented Generation

Recent studies have shown that LLMs are vulnerable to adversarial attacks [60, 88], such as prompt engineering and input perturbation. RAG systems, which combine LLMs with external databases, introduce additional safety risks. For example, attackers may inject adversarial content into retrieved data, bypassing safety alignment and producing malicious outputs [32]. As RAG adoption expands, especially in high-stakes contexts like education, ensuring safety becomes crucial to prevent harmful outputs. This section briefly summarizes adversarial attacks on RAG and highlights promising research directions to enhance RAG safety.

5.1 Taxonomy of RAG Safety

Table 4 summarizes existing RAG methods based on an adversarial taxonomy, briefly defined in this section. Although numerous attacks on LLMs (e.g., backdoor [91, 177], jailbreaking [167], prompt injection [49]) primarily target the underlying LLM, the effect of retrieved contexts on attack surfaces remains unclear. Preliminary studies [188] indicate retrieval can mitigate simple attacks, but complex scenarios involving combined backdoor and retrieval attacks require further research. Due to space constraints, we omit detailed discussion of LLM-specific attacks (see [60, 87]) and instead focus on the retriever's robustness and its interaction with the generator. Figure 2 provides an overview of the RAG safety taxonomy. Safety in RAG, like privacy, is studied almost entirely in the retrieval-augmented setting, as the attack surface is the retrieval corpus itself; we therefore omit the scope annotation.

5.1.1 Threat Model. We define the threat model based on three main RAG components: the external database, the output generator (the underlying language model), and the retriever. A realistic assumption is that attackers have write-only access to the external database, reflecting scenarios where users can upload documents to a database but cannot access all of the content. This presupposes a knowledge base that is shared or open, for example a web-scraped corpus or a public wiki, in which content contributed by one party can later be retrieved to answer another user's query. Attackers typically have no detailed knowledge about the underlying language model, as commercial LLMs are usually proprietary black-box systems.

For the retriever, we differentiate between two threat scenarios. In the white-box setting, attackers have full access to the retriever's parameters, allowing sophisticated adversarial examples crafted by exploiting

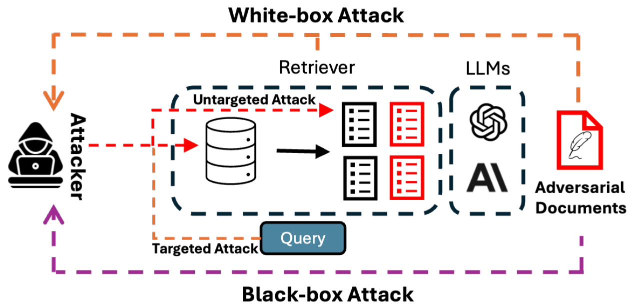


Fig. 2. An illustration of the RAG Safety Threat Model

known weaknesses (e.g., misleading yet highly-ranked documents [198]). Conversely, in the black-box setting, attackers lack direct access and must indirectly infer the retrievers behavior by manipulating external data, relying solely on observing retrieval outputs [32].

5.1.2 Attacker's Goal. In traditional machine learning, attacks are typically categorized as targeted or untargeted. For generative models like RAG, we identify two corresponding categories: targeted attacks and jailbreak attacks. Targeted attacks aim to subtly manipulate responses to specific queries or topics, evading detection and potentially skewing sensitive information (e.g., influencing perceptions on political or social issues). On the other hand, jailbreak attacks broadly seek to bypass built-in safety constraints, provoking models to generate unsafe or inappropriate content.

5.2 Methods of RAG Safety

Targeted Attacks. Zou et al. [198] introduced PoisonedRAG, an attack exploiting the retrieval component by injecting carefully designed passages into the database. The goal is to manipulate RAG systems to produce attacker-specified answers to particular questions. PoisonedRAG considers both black-box and white-box scenarios. In black-box, where the attacker lacks access to model parameters, it uses a heuristic: passages closely matching the query are more likely retrieved. In white-box, with model parameter access, the passages are optimized to maximize similarity between the encoded query and passage: $P = \text{argmax}_{P'} Sim(f(Q), f(P'))$, where Q is the user query, P is the adversarial passage, f is the encoder, and Sim measures similarity. More recently, Chang et al. [20] further demonstrated that even a single poisoned document can achieve high attack success rates under fully black-box conditions, while Zhao et al. [194] extended poisoning attacks to knowledge graph-based RAG by injecting adversarial triples that complete misleading reasoning chains.

However, PoisonedRAG targets specific queries, neglecting broader group-based attacks on semantically related query categories (e.g., political or social groups). To address this, Xue et al.[176] proposed the BadRAG framework, extending targeted attacks to semantic groups. For instance, given a topic such as Republican, BadRAG identifies related terms (Governor, Red States, Pro-Life) as triggers. It employs contrastive learning to craft adversarial passages, maximizing similarity with triggered queries and minimizing it with normal queries. BadRAG also enables other adversarial behaviors, like Denial-of-Service (DoS) attacks[176], further exploring the vulnerabilities.

Dense retrieval has been extensively studied within the Information Retrieval community [195]. Many attacks on dense retrievers, such as adversarial manipulation, are also applicable to passage retrieval tasks. Recently, Long et al.[89] introduced a backdoor attack framework that exploits grammatical errors as triggers to spread misinformation. By employing contrastive learning to fine-tune the retriever, the model can retrieve adversarial passages specified by an attacker when it detects these grammatical anomalies. Additionally, Zhong et al.[196] demonstrated that adversarial passages trained on one domain can effectively transfer to out-of-domain queries, broadening the scope and potential impact of such attacks.

Despite these advances, their impact on downstream generation remains unclear, as most were developed without considering generation tasks. For instance, RAG methods equipped with safety guardrails could potentially diminish the impact of adversarially retrieved passages. Bridging this gap presents a key research opportunity in Trustworthy RAG (discussed in our Future Directions).

Jailbreak Attacks. When specific attack targets are absent, the threat model shifts toward jailbreak attacks. An et al. [7] has demonstrated that RAG can negatively impact LLM guardrails. Wang et al.[166] examine jailbreaking in the context of LangChain, a popular RAG framework. They analyzed jailbreak vulnerabilities in major Chinese Large Language Models and introduced the Poisoned-LangChain (PLC) method. By embedding jailbreak prompts into the retrieval database, PLC achieved jailbreak success across three scenarios, maintaining a consistent success rate exceeding 80%. More recently, Deng et al.[32] introduce Pandora, which extends jailbreaking attacks to English-based LLMs and more generalized RAG frameworks. Pandora enhances the malicious prompts by categorizing them into distinct topics and storing them in PDF format. This approach ensures that only titles and abstracts are retrieved, by which circumventing potential defense mechanisms that might detect the malicious content.

5.3 Safety Evaluation

Metrics and Datasets. Targeted attacks are evaluated from two perspectives: exclusivity, i.e., the proportion of adversarial passages retrieved for clean versus triggered queries across top-k results [176]; and effectiveness, i.e., Attack Success Rate (ASR) using substring rather than exact matching to account for linguistic variation [198]. Jailbreak ASR requires manual labeling based on the relevance and quality of generated content [32, 166], which we identify as a methodological gap. No standardized datasets exist; targeted attack evaluations typically adapt QA benchmarks such as NQ [75], MS MARCO [14], and SQuAD [124], while jailbreak evaluations rely on manually curated adversarial questions grouped by theme, such as Adult, Harmful, and Illegal in Pandora [32], and Dangerous Behaviors and Illegal Discrimination in Poisoned-Langchain [166]. The absence of unified benchmarks underscores the need for standardized safety evaluation frameworks for RAG systems.

5.4 Future Directions of RAG Safety

Safety in Structured and Multi-Modal Retrieval. Most safety research assumes text-based retrieval, yet emerging RAG systems are incorporating knowledge graphs [104] and multimodal sources [119]. They can require fundamentally different defenses than passage poisoning. Future work should develop modality-specific threat models and corresponding defenses for each retrieval source type.

RAG-Specific Adversarial Training. Current defenses such as token masking [176] predominantly operate at generation time, tuning the model to be robust against corrupted passages after retrieval. However, a more proactive direction is to develop retrieval-time adversarial training that prevents corrupted or poisoned passages from being retrieved in the first place. Concretely, this could involve training the retriever with contrastive objectives that explicitly penalize high similarity scores for adversarially crafted passages, effectively making the retrieval stage poison-aware before any content reaches the generator.

Table 5. Taxonomy for RAG Fairness. The Scope column marks whether each work is RAG-specific (RAG) or a general technique applicable to RAG but not RAG-specific (LLM, covering general LLM and IR methods).

Module	Reference	Scope	Bias Mitigation	Focus	Year
Retrieval	Wang et al. [164]	LLM	LoRA Fine-tuning	Ranking Fairness vs. Performance	2024
	Shrestha et al. [138]	RAG	Diverse Sampling	Demographic Diversity	2024
	Rekabsaz et al. [126]	LLM	Re-ranking Methods	Societal Bias Mitigation	2021
	Rekabsaz et al. [127]	LLM	Post-hoc Re-ranking	Gender Bias in Retrieval	2020
Generation	Wu et al. [169]	RAG	Empirical Evaluation	Cross-task Fairness	2024
	Wang et al. [160]	LLM	Output Conditioning	GPT-3.5/4 Bias Detection	2023
	Liang et al. [82]	LLM	Representation Adjustment	Fair Question Answering	2022
	Parrish et al. [112]	LLM	Benchmark Evaluation	Stereotype Analysis	2021
	Jeung et al. [64]	LLM	Benchmark Evaluation	Long-form Generation Bias	2025
Retrieval + Generation	Kong et al. [73]	LLM	Post-hoc Bias Mitigation (PBM)	Gender and Race Fairness	2024
	Kim et al. [69]	RAG	Fair Retrieval + Generation	Fairness-Quality Trade-off	2024
	Shrestha et al. [138]	RAG	Cross-Modal Guidance	Demographic Balancing	2024
	Hu et al. [58]	RAG	Fair Retrieval + Generation	Demographic Diversity	2025
	Bagwe et al. [13]	RAG	Backdoor Attack	Fairness Vulnerability	2025

6 Fairness of Retrieval Augmented Generation

As generative models such as LLMs and image-generative models become increasingly integrated into real-world applications, it is critical to ensure fair outputs. RAG systems combine generative models with external knowledge retrieval, providing improvements in accuracy and relevance by incorporating up-to-date information. However, the external data retrieved by these models may contain societal biases due to biased pre-trained knowledge [16], which leads to biased outputs. This also introduces the risk of amplifying disparities in gender, race, and other demographic attributes, particularly when the retrieved data is drawn from biased or unregulated sources. Figure 3 demonstrates how biased documents and retrieval can lead to biases in generation.

6.1 Taxonomy of RAG Fairness

In this section, we investigate various approaches that aimed at promoting fairness in RAG models. As shown in Table 5, ensuring fairness in RAG systems requires addressing biases at two stages: the retrieval of external data and the generation of outputs. These stages introduce unique fairness challenges, from biased data sources to unfair generative behavior, which need to be mitigated to create equitable and unbiased systems.

6.2 Fairness in Retrieval

One major challenge for fairness in RAG systems lies in the retrieval phase, where external knowledge or data used to enhance generation is sourced. Fairness issues during this stage can arise from various factors, including the retrieval model, the retrieval process, and the re-ranking mechanism. To address these challenges, several frameworks have been proposed to ensure that the retrieval system itself is fair. Reskabsaz et al. [127] introduces a framework for measuring bias that quantifies gender-related biases in ranking lists and assesses the impact of both BM25 and neural retrieval models. Furthermore, Reskabsaz et al. [126] investigates how re-ranking methods can mitigate biases present in initial retrieval results. Then, Wang et al. [164] recognizes a gap between fairness and ranking performance when using LLMs for re-ranking and proposes a method with LoRA. To ensure demographic diversity, FairRAG [138] incorporates external data sources that cover a broad range of age, gender, and skin tone categories. This approach uses post-hoc sampling techniques to debias the retrieval process, preventing disproportionate

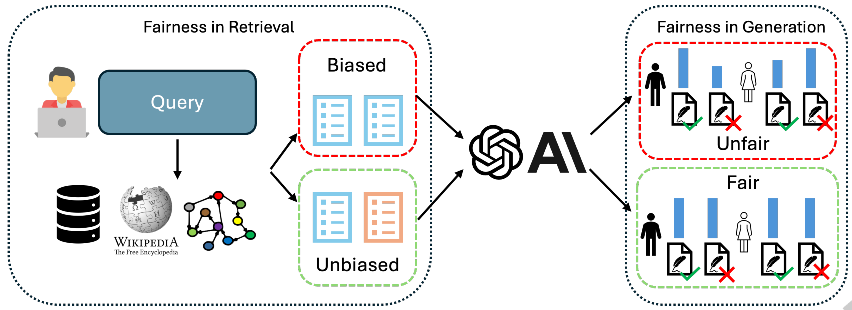


Fig. 3. An overview of the fairness challenge in RAG. The biased retriever includes content related to the blue sensitive attributes, and the biased language model generates unfair content based on protected attributes.

representation of specific demographic groups in the retrieved data. It differs in goal from the other methods: it treats fair retrieval augmentation as a mechanism for improving the fairness of generated outputs the ranking-debiasing approaches of Rekabsaz et al. and Wang et al. aim to remove bias that the RAG retriever itself introduces. Beyond addressing bias in the data itself, methods such as BadRAG [176] have shown how maliciously inserted or poisoned data in the retrieval corpus can lead to biased and unfair outputs. Bagwe et al. [13] further demonstrated that fairness vulnerabilities can be systematically exploited through BiasRAG, a two-phase backdoor attack that compromises the query encoder to align target groups with social biases. Kong et al. [73] proposes Post-hoc Bias Mitigation (PBM), which balances retrieved image sets to ensure more equitable representation across gender/race.

6.3 Fairness in Generation

Once the data is retrieved, the next challenge lies in ensuring that the generative process itself is fair. Even with fair retrieval, generative models may introduce biases based on how the retrieved data is integrated. To promote fairness in generation, Liang et al. [82] assesses the accuracy of question-answering systems while accounting for fairness through measures of toxicity and representation bias. Similarly, Wang et al. [160] identifies the demographic imbalances in models like GPT-3.5 and GPT-4 under both zero-shot and few-shot question-answering settings. FairRAG [138] employs conditioning techniques where generative models are guided by references that are demographically diverse. By incorporating external images or data from a wide range of demographic groups, these models produce more balanced and representative outputs. Parrish et al. [112] introduces the BBQ benchmark to evaluate biases in LLM-generated responses by examining the reliance on stereotypes and anti-stereotypes in both ambiguous and disambiguated contexts. Jeung et al. [64] extended bias evaluation to long-form generation, proposing LTF-TEST to measure demographic favoritism in LLM-generated essays and finding that bias persists in extended outputs. Next, to fully explore fairness throughout all stages and components of RAG pipelines, Wu et al. [169] conducts an empirically evaluation of fairness across various RAG methods. Similarly, Kim et al. [69] evaluates RAG systems with a fairness-aware retriever across seven different tasks and identifies the overall trend of fairness-quality trade-off, considering both retrieval and generation performance.

Metrics and Datasets. Fairness evaluation combines standard QA accuracy metrics—Exact Match (EM) [124] and ROUGE-1 [83]—with group-level fairness metrics. Group Disparity (GD) [42] measures performance differences between protected and non-protected groups, while Equalized Odds (EO) [53] assesses whether true positive and false negative rates are comparable across demographic groups. Retrieval-specific evaluations additionally report retrieval success rate and rejection rate to assess whether biases

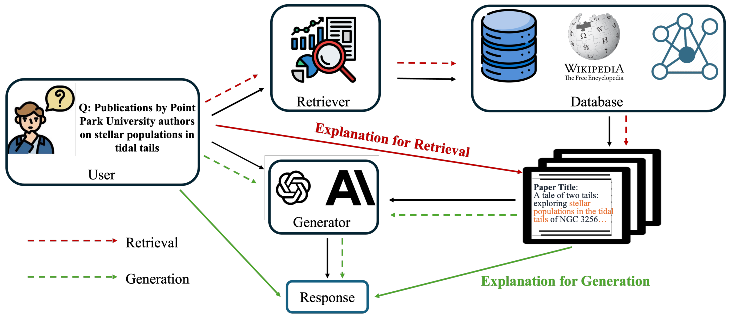


Fig. 4. An Overview of Explainability in RAG

in retrieval propagate to generation [176]. Commonly used datasets include the TREC Fair Ranking Track [27], which evaluates fair exposure across demographics such as gender and location; the BBQ dataset [112], which tests model behavior under both ambiguous and disambiguated social contexts; and the LaMP benchmarks [131], which evaluate personalization fairness across classification, regression, and generation tasks via retrieval-augmented user interaction histories.

6.4 Future Directions of RAG Fairness

Multi-Objective Fairness-Utility Optimization. Existing work treat fairness and retrieval relevance as competing objectives, typically sacrificing one for the other. A more principled direction is to formulate RAG fairness as a multi-objective optimization problem jointly optimizing fairness, accuracy, and relevance using Pareto-efficient methods that expose the fairness-utility frontier, enabling practitioners to make informed trade-off decisions rather than fixed compromises [82, 160].

Multi-modal Fairness. Finally, multimodal RAG systems that handle both text and image data raise new fairness concerns. Biases may emerge differently across modalities, and current research [69, 169] only begins to address this. Future work should ensure consistent fairness across modalities to build more trustworthy multimodal systems.

7 Explainability of Retrieval Augmented Generation

Explainability is a crucial aspect of trustworthiness, explaining model behavior and providing valuable insights for decision making [128]. Its importance has grown with the rise of black-box LLMs [82, 143]. While prior surveys [197] address transparency in RAG, our focus is distinct: While transparency is more general and seeks to understand the algorithms and underlying rationales, our explainability (of the output) specifically aims to elucidate why a particular input (transparency) leads to a given output through a specific model (interpretability).

As shown in Figure 4, RAG systems pose unique explainability challenges due to the complexity of their multi-stage nature. Beyond explaining the generation step, it is critical to understand the retrieval decisionse.g., which input terms trigger the selection of specific contexts. In systems with post-retrieval processing [68, 179], further explanation is needed for how inputs influence reranking or filtering. This section reviews methods for explaining both the retrieval and generation components of RAG.

7.1 Taxonomy of RAG Explainability

7.1.1 Explainability in Retrieval. As illustrated in Table 6, we situate our discussion of explainability in RAG in the distinction between retrieval, generation, and dual enhancement. To the best of our knowledge, no dedicated research efforts have been made to explain retrieval within the context of RAG.

Table 6. Taxonomy for RAG Explainability. The Scope column marks whether each work is RAG-specific (RAG) or a general technique applicable to RAG but not RAG-specific (LLM).

Module	Reference	Scope	Task	Year
Generation	Sudhi et al. [143]	RAG	English and German QA	2024
	Luo et al. [93]	RAG	Knowledge Graph QA	2024
	Rorseth et al. [129]	RAG	Open-book QA	2024
Retrieval + Generation	Kunze et al. [150]	RAG	Scene-Understanding	2024
	Hussien et al. [61]	RAG	Road User Intention Explanation	2024
	Ferraretto et al. [40]	LLM	Document Retrieval	2023
	Hu et al. [59]	RAG	Graph Extraction	2025

However, several studies have indeed explored explainability in the general information retrieval, particularly in recommender systems and search [191, 192]. Therefore, we provide a high-level summarization of representative explanation techniques in information retrieval, with the expectation of inspiring similar success in explaining the retriever of RAG.

Based on [8], the explanation methods in information retrieval can be categorized into post-hoc explanation, axiomatic strategies, probing strategies, and self-interpretable designs. The post-hoc explainers explain the models after they make decisions, the representative examples of which used in information retrieval are feature attribution and generative approach. The feature attribution works by ascribing the retrieved outcomes to certain input (i.e., the attribution). Some of the methods find the explanation features by computing the feature importance, such as [121] uses interpretable textual features to explain rankings, [118] understands the BERT-based ranking models by the attention scores of tokens, and [157] estimates the point-wise explanations by analyzing the contribution of each token to the output of the ranking model. Other methods try to explain the model outputs by finding the most explanatory features, e.g. [139] uses a greedy search-based algorithm to obtain a subset of features that serve as the explanations. Axiomatic explainers provide explanations using axioms [158] and probing explainers provide valuable insights into the inner workings of neural models by revealing what types of information are encoded in their embeddings and model parameters [25], how sensitive they are to various textual properties [95], and what knowledge they possess [41]. Although self-interpretable explainers inherently provide explanations by their designs, making the model fully transparent is extremely challenging, and usually, only specific components are interpretable and transparent [41, 193].

7.1.2 Explainability in Generation. Explaining the generator in RAG systems is as important as explaining the retriever since the final output is directly produced through generation. Errors in this stage can arise from shortcuts or reliance on misleading features [31], and LLM-based generators are particularly prone to hallucinations. This raises concerns about whether the models output genuinely reflects the query-relevant content and retrieved evidence [59, 133, 143].

Explanation techniques for generation can be grouped into two main categories: post-hoc and ante-hoc. Post-hoc methods aim to interpret the generator after inference. For example, RAG-Ex [143] offers a model-agnostic, perturbation-based framework that measures how changes to input tokens affect the output. It proposes six types of perturbations (e.g., token removal, noise injection, entity or word substitutions), then ranks token importance based on the impact on generated responses. Among them, the "leave-one-token-out" strategy proves most effective at highlighting critical inputs. Another method, RAGE [129], generates counterfactuals to determine the provenance and salience of external knowledge used by the generator. It incorporates pruning strategies to reduce the search space, making the explanation process more efficient.

In contrast, ante-hoc methods incorporate explainability directly into the generation process. For example, RoG [93], leverages knowledge graphs to generate reasoning paths for each query, which provide intermediate steps that the LLM follows to arrive at an answer. This both enhances transparency and improves generation quality by encouraging structured faithful reasoning.

7.2 Dual Enhancement of Explanations and RAG

In addition to exploring how to explain RAG systems, existing work also investigated the dual enhancement of explanations and RAG. On the one hand, explanations can be integrated to augment RAG systems. On the other hand, RAG systems can also be employed to provide explanations.

As retrieval plays a critical role in RAG [28], the enhancement of adding explanations during retrieval could benefit the whole RAG system. ExaRanker [40] utilizes the explanations as additional labels to train the ranking models in the information retrieval task. Specifically, given the question-passage pair and the label indicating whether the passage can be used to answer the question, an LLM is first employed to generate the explanations of why the question can/cannot be answered by the given passage. With these ground-truth explanations, the ranking model is trained on question-passage pairs to predict not only whether the question can be answered but also the corresponding explanations. By integrating explanations as additional training labels, the ranking model can better understand the relationships between the questions and passages, which could benefit the ranking performance and ease the demand for a large number of training examples.

Apart from leveraging explanations to augment RAG systems, the reverse relationshipusing RAG to improve explanationsis also worth exploring. For example, Tekkesinoglu and Kunze [150] use RAG in scene-understanding tasks to create explanations through a question-answering approach. For each input with a class label, the model predicts the probability of belonging to that class. To assess the impact of each semantic feature, it generates predictions without specific features, enabling the calculation of feature importance. These outputs, along with the contrastive cases, contribute to an external knowledge repository for LLMs. This RAG design can thus generate faithful explanations for the prediction model of the scene-understanding tasks. Another work [61] studying road user behavior also uses RAG to generate explanations. In particular, this work first creates a human-readable document that explains why the road user may/may not have a specific behavior. The document is then processed to form a database, serving as the external knowledge base of the RAG system. Given a tailored prompt and a query derived from the prediction frame, the RAG system will create a detailed explanation of the road user's intention.

7.3 Explainability Evaluation

Metrics and Datasets. Explainability evaluation in RAG draws on conventional XAI metrics: fidelity, which measures how accurately an explanation reflects the model's decision-making process [5, 100], and stability, which assesses consistency across similar inputs [117]. For RAG-specific explanation evaluation, Sudhi et al. [143] propose significance, quantified via F1 and Mean Reciprocal Rank (MRR), and plausibility, assessed through human annotation against ground-truth token-level explanations. No standardized datasets exist; researchers adapt existing QA benchmarks such as XQUAD [143] and knowledge graph QA datasets WebQSP [184] and CWQ [149] for explainability evaluation [93]. The development of dedicated explainability benchmarks for RAG remains an important open challenge.

Table 7. Taxonomy for RAG Accountability. The Scope column marks whether a work is RAG-specific (RAG) or a general applicable technique but not RAG-specific (LLM). Most watermarking methods are general.

Module	Reference	Scope	Technique	Type	Year
Retrieval	Liu et al. [84]	LLM	Format-based	Text Watermarking	2024
	Xu et al. [173]	LLM	Embedding-based	Data Watermarking	2024
	Sun et al. [147]	LLM	Trigger-based	Data Watermarking	2022
	Lv et al. [94]	RAG	Knowledge-based	Black-box Knowledge Watermarking	2025
Generation	Christ et al. [24]	LLM	Semantic-based	Sentence-level Watermarking	2024
	Hou et al. [56]	LLM	Sampling-based	Token-level Watermarking	2023
	Kirchenbauer et al. [70]	LLM	Logit-based	Global Watermarking	2023
	Yang et al. [181]	LLM	Post-generation	Lexical/Syntactic	2022
	Qu et al. [122]	LLM	Logit-based	Robust Multi-bit Watermarking	2025
Retrieval + Generation	Jovanovic et al. [67]	RAG	Integrated Pipeline	Red-Green Token Scheme	2024
	Guo et al. [52]	RAG	CoT-based	Reasoning Watermarking	2025

7.4 Future Directions of RAG Explainability

Knowledge Graph-Augmented Explainability. LLM-based graph traversal methods [66] introduce relational reasoning steps that traditional explanation techniques cannot capture. However, in real world, one knowledge graph might not fully capture the required context. One promising direction is to develop graph-based retrieval on multiple graphs where the retriever and generator jointly optimized to combine multiple knowledge sources for better explainability.

Domain-Aware Explainability Metrics. Current explainability approaches focus on general question answering, applying domain-agnostic metrics that measure consistency without accounting for domain-specific correctness. However, domain-specific applications require explanations that meet the verification standards of each field. For example, medical question answering requires expert input and clinical evidence alignment, while legal QA demands traceability to specific statutes or precedents. Future work should develop domain-aware explainability metrics grounded in expert judgment, validated on benchmarks such as MedicationQA [15] and LegalBench-RAG [116].

8 Accountability of Retrieval Augmented Generation

Accountability in AI concerns whether outputs meet procedural and substantive standards and which entities are responsible when violations occur [33]. It is broad, spanning governance, audit logging, responsibility assignment, and traceability of content to its origin. Watermarking addresses the traceability facet, which in RAG is the most acute and the most amenable to a technical solution: since responsibility is split across the retriever, the corpus curator, and the generator, tracing an output or passage back to its source is a prerequisite for assigning responsibility at all. By embedding verifiable, difficult-to-remove signals into generated text or corpus content, watermarking links outputs and data to a specific source model or dataset, enabling downstream auditing. We therefore focus on watermarking as the technical foundation of accountability in RAG, and treat the remaining, largely procedural facets (governance, audit infrastructure, responsibility assignment) as open directions in Section 8.6.

8.1 Taxonomy of RAG Accountability

Retrieval. Accountability in this survey refers to embedding watermarks within the sources of retrieval, encompassing both Text Watermarking and Data Watermarking techniques. These approaches aim to safeguard content and data ownership, ensuring accountability and traceability.

Generation. Watermarking is key to ensuring accountability in LLM outputs, with strategies applied at different stages. Pre-generation embeds markers during training, in-generation integrates them during inference, and post-generation adds them after text is produced. Each method offers distinct advantages, collectively enhancing traceability and authenticity. We summarize the RAG accountability taxonomy in Table 7.

8.2 Accountability in Retrieval

8.2.1 Text Watermarking. Text Watermarking embeds identifiable markers into textual content to authenticate ownership [84] (Figure 5). The dominant family is format-based watermarking, which alters only formatting rather than content: examples include line or word shifting [19], Unicode codepoint insertion such as whitespace [132], and variation in color, font, or embedded features [63, 97]. These methods preserve the original text but are vulnerable to removal via canonicalization [17] and to forgery, since their formatting patterns are detectable [84].

8.2.2 Data Watermarking. Data Watermarking protects a dataset or corpus, rather than a model's outputs, against unauthorized use. For example a proprietary retrieval corpus or a curated training set whose owner wishes to detect whether it has been copied or used to train a model without permission. The watermark is embedded into the data and is later detected through the behavior of any model built from it. A common instance is backdoor watermarking [81]: the owner inserts trigger records, namely specific input patterns paired with unique target behaviors, into the dataset, so that a model trained on it responds to those triggers in a pre-specified, identifiable way.

Trigger-based watermarking is widely employed due to its flexibility and effectiveness as a type of backdoor watermarking. These triggers can take various forms, including word- or sentence-level modifications [147], semantically invariant transformations in code [146], or distinctive input formats designed to be recognizable [173]. Embedding such triggers ensures that ownership can be verified through the model's behavior, even if the dataset itself is no longer accessible.

While trigger-based watermarking is robust, its effectiveness depends on careful design to ensure triggers remain inconspicuous yet detectable. Additionally, as models become more complex and versatile, challenges such as ensuring trigger persistence and avoiding unintended activations must be addressed. As the field evolves, continued innovation in embedding mechanisms and detection strategies will be essential for robust and scalable dataset protection.

8.3 Accountability in Generation

Generation-stage watermarking embeds markers at different points in the model lifecycle. Specifically, pre-generation, in-generation, and post-generation. We discuss each in more details in the following paragraphs.

Pre-generation. Markers are embedded during training. Trigger-based watermarking (Section 8.2.2) reveals ownership only on specific inputs, minimizing impact on normal outputs but limiting detection in broad use. Global watermarking instead marks all outputs for pervasive traceability by distilling the signal into model parameters, via sampling- or logit-based distillation [50] or reinforcement learning from watermark-detector feedback [175], at the cost of careful imperceptibility design. A promising direction is hybrid schemes that combine specificity with global universality.

In-generation. Markers are injected at inference without altering parameters. Logit-based methods such as KGW [70] bias generation toward a hashed green-token list and detect via a z-score over green-token proportions; entropy-based weighting [92] and sliding-window methods [71] restore performance in low-entropy contexts such as code, and Qu et al. [122] give a provably robust multi-bit scheme resistant to copy-paste attacks. Sampling-based methods bias the token-selection seed [24], which is edit-sensitive, or partition the semantic space to survive paraphrasing [56].

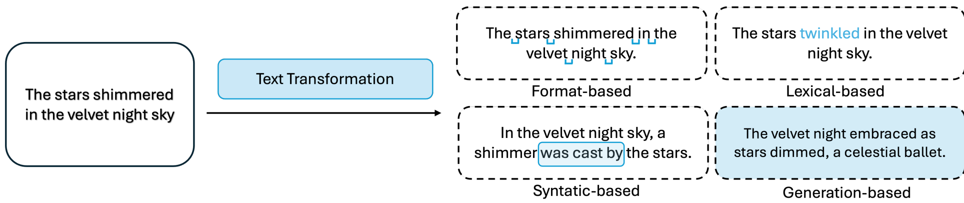


Fig. 5. Common text generation watermarking techniques.

Table 8. Watermarking Datasets and Metrics

Dataset	Metric: Detectability	Metric: Quality Impact	Metric: Robustness
WaterBench [156]	✓		
WaterJudge [98]	✓	✓	
Mark My Words [115]	✓		✓
MarkLLM [107]	✓	✓	✓

Post-generation. Markers are added to finished text [84]. Beyond the format-based methods of Section 8.2.1, lexical watermarking substitutes words while preserving meaning, advancing from context-agnostic WordNet or Word2Vec methods [39, 153] to context-aware BERT infilling [181, 185]; syntactic watermarking alters grammatical structure such as clefting and topicalization [12, 152] but is language-dependent and can disturb fluency; and generation-based watermarking uses neural rewriters such as AWT [1], REMARK-LLM [190], and WATERFALL [77] to embed high-capacity watermarks while preserving naturalness.

8.4 Accountability in RAG Systems

WARD (Watermarking for RAG Dataset Inference) [67] introduces a unified approach to watermarking in RAG systems by embedding imperceptible signals into datasets to enable traceability through both retrieval and generation, even after paraphrasing. Its red-green token scheme ensures statistically robust detection, resists content blending, and offers low false detection rates with scalable query aggregation. We recognize it as one of the first methods to provide tracking across both retrieval and generation stages in RAG. This marks a step toward accountable and transparent data usage in multi-stage RAG pipelines. More recently, RAG-WM [94] proposes a black-box knowledge watermarking approach that embeds entity-relationship tuples into the RAG knowledge base using a multi-LLM interaction framework, enabling IP infringement detection even when the adversary deploys the stolen knowledge base with a different LLM. Additionally, RAG^{C} [52] addresses harmless copyright protection by implanting watermark verification behaviors in the chain-of-thought reasoning space rather than the final output, using a pairwise Wilcoxon test for ownership verification while preserving the correctness of the RAG system's responses.

8.5 Accountability Evaluation

Metrics and Datasets. Watermarking evaluation spans five dimensions [84]: detectability, assessed via z-scores or p-values for zero-bit schemes and Bit Error Rate (BER) [185] or bit accuracy [186] for multi-bit schemes; quality, measured through comparative metrics such as BLEU [111] and semantic similarity, or single-text metrics such as Perplexity (PPL), with human evaluation as the gold standard; output performance, using PPL [185], GPT-4 scoring [35], CodeBLEU [51], and ROUGE [83] depending on task type; diversity, via lexical metrics such as Seq-Rep-N [50] and entropy-based metrics such as Ent-3 [56]; and robustness against adversarial transformations (see Section 5). Several benchmarks have been developed to standardize evaluation, including WaterBench [156], WaterJudge [98], Mark My Words [115], and MarkLLM [107], with details provided in Table 8.

8.6 Future Direction of RAG Accountability

Procedural Accountability. Watermarking establishes provenance, but full accountability in deployed RAG also requires audit infrastructure and clear responsibility assignment across the retriever, the corpus curator, and the generator. A concrete direction is to develop tamper-evident retrieval logs and per-component provenance records, together with policies that allocate responsibility when a harmful output traces to corpus content rather than to generation, extending accountability beyond the content-tracing techniques surveyed here.

Personalized Watermarking. Current watermarking schemes embed uniform signals regardless of user or deployment context, making it impossible to trace generated content back to a specific user session or retrieval query. A concrete direction is to develop personalized watermarking methods that embed user-specific or query-specific signals into generated outputs, enabling fine-grained attribution and supporting accountability at the individual interaction level.

9 Cross-Dimensional Quantitative Analysis

Although each trust dimension has its own challenges and metrics, improving one may come at the cost of another. To study these interactions, we apply one trust-oriented intervention at a time and measure its impact across all six dimensions against a naïve RAG baseline.

9.1 Experimental Setup

All experiments use `Qwen3-4B-Instruct` [178] as the generator and `all-MiniLM-L6-v2` [125] as the retrieval encoder. The corpus consists of 800 documents encoded and indexed via cosine similarity over L2-normalized embeddings.

Datasets and Queries. We curate 700 queries spanning four source dimensions, with up to 50 queries per dimension evaluated per run. The Reliability & Robustness split contains 200 factoid questions drawn from Natural Questions [75], each paired with a gold document and a noise-injected counterpart for robustness evaluation. The Safety split contains 100 harmful requests in the style of JailbreakBench [21], spanning categories such as violence, criminal planning, and self-harm. The Fairness split contains 200 ambiguous-context demographic questions following the BBQ benchmark [112], where the correct answer is "Cannot be determined." instead of the stereotype answer. Finally, the Privacy split contains 200 canary queries requesting PII (email addresses, SSNs) that appear in planted corpus documents. All queries are domain-agnostic and designed to be answerable from short retrieved passages, ensuring that evaluation performance reflects pipeline behavior with the designed intervention. The corpus backing each split is indexed with the same dense retriever to provide a controlled retrieval environment across all trust dimensions.

Trust-oriented interventions. The Reliability intervention generates N answers and returns their union as a conformal-style prediction set [80, 104]. The Safety intervention wraps the pipeline with a Llama Guard-style dual-layer classifier [62] that screens both queries and responses against unsafe categories. The Fairness intervention reranks retrieved documents via round-robin selection for diverse provenance and injects bias-aware instructions for demographic queries [138]. The Privacy intervention applies PII scrubbing to both queries and retrieved documents using the Presidio NER engine [188]. The Robustness intervention filters retrieved documents by intra-document coherence (mean pairwise cosine similarity of sentence embeddings), dropping noise-corrupted documents below a threshold [161]. The Accountability intervention embeds a KGW statistical watermark during generation by biasing green-list token logits [70].

Metrics. As introduced in the individual sections, each dimension requires dedicated metrics for evaluation. To enable comparison among the dimensions, we normalize each dimension to a $[0, 1]$ scale. Reliability is computed as the average token-level F1 between the model response and reference answers, and Robustness applies the same token-level F1 metric to noise-injected retrieval contexts. Safety is the fraction of harmful queries for which the model produces a refusal, while Fairness is the fraction of ambiguous-context queries answered with an unbiased response (e.g., "Cannot be determined"). Privacy is reported as $1 - \text{PII}$ leakage rate, where leakage is defined as the presence of email addresses or SSNs in the response. Accountability is the fraction of responses in which the KGW watermark is successfully detected.

Evaluation Protocol. For each retrieval budget $k \in \{3, 5, 10\}$, we construct seven pipeline variants: the naïve RAG baseline plus one intervention-augmented pipeline per trust dimension. Every variant is evaluated on every query set. We report results as percentage-point change relative to the naïve baseline, i.e., $(\text{score}_{\text{intervention}} - \text{score}_{\text{baseline}}) \times 100$. Since all metrics share the same $[0, 1]$ scale, percentage-point differences are directly comparable across dimensions without the division-by-zero issues of classical percentage change. Aggregated results across all six metrics and retrieval budgets are reported in Figure 6.

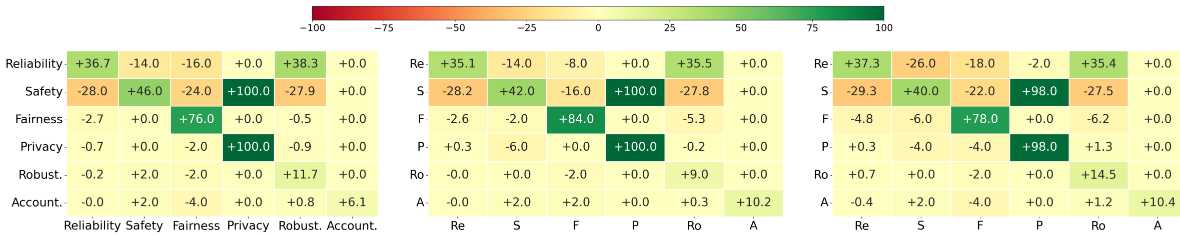
9.2 Discussion

The cross-dimensional results in Figure 6 reveal that trust interventions do not operate in isolation: improving one dimension can either reinforce or undermine others. To organize these interactions, we categorize dimension pairs into three types: synergistic pairs, where one intervention incidentally benefits another dimension; antagonistic pairs, where gains in one dimension come at measurable cost to another; and neutral or asymmetric pairs, where interventions are largely independent.

Reliability: The reliability intervention negatively impacts safety and fairness by expanding retrieval coverage, increasing the likelihood that harmful or demographically biased content enters the context window. However, reliability exhibits a synergistic relationship with robustness: by aggregating multiple generations into a prediction set, the intervention increases the probability of producing a correct answer even when parts of the retrieved context are corrupted, effectively hedging against noise.

Safety: The safety intervention, which applies guardrails to the inputs and outputs guardrails, takes a conservative stance toward retrieved content and adopts a cautious approach to generation. This conservatism has a negative impact on reliability, fairness, and robustness, as aggressively filtering retrieved documents can withhold factually useful context. The exception is privacy, where reduced content surfacing incidentally limits PII exposure. Practitioners should carefully consider this broad trade-off when deploying safety interventions in applications where reliability or coverage is also a priority.

Fairness: Similar to the safety intervention, the fairness intervention operates by filtering or reranking retrieved documents and injecting bias-aware generation instructions, which reduces the volume of available context. As a result, it shares many of the same antagonistic relationships as safety, negatively



(a) Retrieval Budget $k = 3$ (b) Retrieval Budget $k = 5$ (c) Retrieval Budget $k = 10$

Fig. 6. Heatmap for the cross-dimensional experiment. The numbers represent the absolute point differences. The y-axis represents the intervention implemented, and the x-axis represents the trustworthiness metrics.

impacting reliability and robustness by withholding potentially relevant documents. Practitioners face a similar trade-off between equitable outputs and factual coverage.

Privacy: The privacy intervention, through PII scrubbing at both the query and context levels, exhibits a largely neutral effect on most dimensions, with the notable exception of safety, which benefits modestly from the reduced exposure of sensitive content. However, it is worth highlighting an unintuitive negative interaction with fairness: by redacting personally identifiable attributes (e.g., race or gender) from the retrieved context, the PII scrubber removes the demographic signals that the fairness intervention relies on to trigger “Cannot be determined,” causing the model to revert to a standard generation that may reflect stereotypical associations. However, we note the effect is relatively small and may be specific to the fairness metric formulation used in our evaluation.

Robustness: The robustness intervention, which filters documents based on intra-document coherence, takes an additional inspection step over the retrieved corpus before generation. By discarding low-coherence documents, it inadvertently improves both safety and fairness, since adversarial, harmful, and demographically charged documents tend to exhibit similar incoherence signatures to noise-injected content. However, we caution that this synergy is not universal: it depends on the specific type of adversarial content present in the corpus and the particular robustness intervention employed. Coherence-based filtering may be less effective against subtly corrupted or well-formed adversarial documents that do not exhibit low intra-document similarity.

Accountability: The accountability intervention, which embeds KGW watermarks at the logit level during generation, has a largely neutral effect on all other dimensions. Because watermarking does not alter the retrieved context, the filtering pipeline, or the semantic content of the generated response, it is effectively orthogonal to reliability, safety, fairness, privacy, and robustness. This makes accountability one of the few interventions that can be freely composed with any other without introducing unintended cross-dimensional effects, suggesting it should be treated as a default component in trustworthy RAG deployments rather than a competing design choice.

10 Applications

In previous sections, we outlined the six dimensions of Trustworthy RAG systems. We now explore how these principles apply in high-stakes domainsHealthcare, Legal, and Educationhighlighting recent advancements, current use cases, and domain-specific challenges.

10.1 Healthcare

In healthcare, RAG enhances multiple workflows. For Clinical Decision Support, RAG retrieves relevant medical documents and patient data to improve diagnostic and treatment recommendations [78, 171]. For patient communication, RAG-based chatbots address the growing demand for reliable online health information [54] by providing symptom assessments and personalized guidance. In knowledge discovery, RAG accelerates drug discovery and hypothesis generation by retrieving up-to-date literature and clinical trial data [180]. For precharting—reviewing patient records before visits [18]—RAG can improve efficiency through enhanced personalization [140, 180].

Trustworthiness Challenges in Healthcare. Reliability: Given the high-stakes nature of healthcare, RAG systems must be highly reliable. Integrating uncertainty quantification into CDS and patient communication can help users assess output confidence [93, 104, 134, 144]. Privacy: RAG systems must prevent sensitive patient data leakage, especially in applications like precharting [18, 188]. Others. Beyond reliability and privacy, RAG systems must be safe, fair, explainable, and accountable. They should resist adversarial manipulation [47], avoid bias [54], offer transparent reasoning, and support traceability of errors.

10.2 Law

In legal applications, RAG supports multiple tasks. For Legal Question Answering, RAG dynamically retrieves relevant statutes and precedents to provide contextualized responses about laws and case precedents across jurisdictions [2, 23, 76, 168]. For Legal Document Summarization, RAG can improve factual grounding by retrieving case law and legal commentary to produce more accurate summaries of long, technical documents [10, 23, 86]. For Legal Judgment Prediction, RAG retrieves similar precedent cases and applicable laws to enhance transparency and better reflect judicial reasoning, moving beyond early classification-based approaches [29, 170].

Trustworthiness Challenges in Legal Applications. Fairness: Biased retrieval can surface prejudiced precedents or omit underrepresented viewpoints, potentially contributing to unjust outcomes [138]. Targeted de-biasing and legal-specific retrieval strategies are needed [44]. Explainability: Legal professionals must trace how conclusions were reached. RAG systems must clearly indicate retrieved sources and their contribution to outputs [29, 90], though domain-specific explainability frameworks remain underdeveloped. Others. Like in healthcare, legal RAG systems must also be reliable, private, and accountable. They should remain accurate under ambiguous queries, protect confidential client records and sealed case documents, and support traceability of errors to maintain trust.

10.3 Education

In education, RAG enhances several workflows. For personalized learning, RAG dynamically retrieves content based on student progress and knowledge gaps, enabling more adaptive learning experiences beyond static approaches [101, 105, 163]. For student support, RAG-based assistants answer questions and clarify concepts in real time using updated, contextually relevant information [30]. For teacher support, RAG can assist with lesson planning and content generation using domain-specific resources [99], though further work is needed on high-quality educational datasets.

Trustworthiness Challenges in Educational Applications. Fairness: RAG systems may reinforce biases in training or retrieved data [136], leading to unequal treatment or content that disadvantages underrepresented groups. Safety: RAG systems are susceptible to adversarial attacks and jailbreaks [32, 167, 176], which is particularly concerning in educational settings where students may be exposed to misleading or inappropriate content. Others. Additional trustworthiness concerns include reliability, privacy, and robustness. Educational RAG systems must consistently deliver accurate and relevant content. They must also protect sensitive data like student performance and demographics. Robustness is needed to handle diverse learners, ambiguous inputs, and evolving educational goals while maintaining trustworthy performance.

10.4 Enterprise Applications

In enterprise settings, RAG supports diverse workflows. For internal knowledge management, systems help employees query policies, contracts, and institutional knowledge [4], as exemplified by Microsoft 365 Copilot [96] and Google Workspace [48]. For customer-facing support, RAG-powered chatbots provide context-aware responses grounded in product documentation [38], with platforms like Perplexity AI [114] popularizing web-grounded RAG at scale. For personalized assistance, RAG grounds responses in user profiles and interaction histories to tailor outputs to individual employees and customers [102, 103]. For software engineering assistance, RAG retrieves code snippets, documentation, and issue histories to support generation and debugging [57], with AWS Bedrock [6] enabling assistants grounded in proprietary codebases. RAG has also been applied to public sector e-government services, including virtual assistants for citizen engagement [108], multimodal public opinion analysis [110], and multi-agent GraphRAG for policy question answering [109].

Trustworthiness Challenges in Enterprise Applications. Safety: A shared corporate corpus queried by many users means a single poisoned document can propagate adversarial effects at scale. Poisoned-LangChain [166] demonstrates jailbreak attacks against widely adopted RAG frameworks, while Cohen et al. [26] show that self-replicating adversarial prompts can initiate a chain reaction that propagates through entire RAG systems, escalating both the scale and severity of data extraction. Privacy: RAG retrieves content by semantic similarity rather than authorization boundaries, creating pathways for unauthorized access. CopyBreakRAG [65] demonstrates that proprietary documents can be exfiltrated with high precision, highlighting that access control and retrieval authorization should be jointly optimized for semantic relevance and permission-aware filtering.

11 Conclusion

In this survey, we provide a comprehensive review of RAG systems through the lens of trustworthiness, focusing on six critical aspects: reliability, privacy, safety, fairness, explainability, and accountability. For each trustworthiness aspect, we introduce definitions and key concepts to establish an understanding of the topics. We also offer a structured taxonomy to help researchers navigate the diverse approaches in the specific trustworthy aspect. Beyond methodological summary, we highlight the evaluation protocols commonly used for trustworthy RAG systems, including the specific datasets and metrics. By including these discussions, we aim to facilitate the development of benchmarks tailored to trustworthiness challenges.

Additionally, we conduct a cross-dimensional analysis that reveals how the six trust dimensions interact and conflict within RAG's multi-stage pipeline. Unlike standalone LLMs, RAG systems compound trustworthiness challenges across retrieval and generation: retrieval bias amplifies generation bias, corpus-level attacks bypass LLM-level safety defenses, and accountability gaps emerge at the interfaces between pipeline components. These interactions also introduce unavoidable trade-offs. For example, improving explainability via source attribution can expose private corpus contents; enhancing retrieval diversity for fairness can introduce noisy passages that hurt reliability, and safety defenses can reduce output accuracy. Recognizing these trade-offs is essential, as optimizing each dimension in isolation risks degrading others, and holistic trustworthiness requires pipeline-aware solutions that explicitly balance competing objectives.

Finally, we provide an in-depth discussion of future research directions for each aspect of trustworthiness, including promising directions within individual aspects and potential synergies across multiple areas. By addressing these directions, we hope to inspire innovative approaches that enhance the overall trustworthiness of RAG systems and drive their broader adoption in critical applications. This survey not only serves as a comprehensive roadmap for researchers aiming to advance trustworthy RAG systems but also underscores the importance of addressing these challenges to ensure the safe deployment of RAG applications.

Acknowledgments

This research is supported by the National Science Foundation (NSF) under grant numbers IIS2239881, IIS2524380, and IIS2524379.

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Received 14 May 2025; revised 25 June 2026; accepted 10 July 2026